Project 100: Comprehensive Smart Campus Management Simulation

Python Mini Project - College Assignment

1. Aim

Develop a menu-driven Python program to integrate student records, attendance, semester marks, event registration, resource allocation, and campus summary reporting.

2. Algorithm

1. Start and initialize student records, attendance, marks, events, and resources.

2. Display the main menu and read the user choice.

3. Use conditional statements to perform the chosen task.

4. Display student records, attendance eligibility, or marks totals and averages.

5. Register participants for events or allocate available resources.

6. Display a summary of attendance, event registrations, and remaining resources.

7. Repeat the menu until Exit is selected; then stop.

3. Complete Python Program

```
# Project 100: Smart Campus Management System

print("====================================")
print("      SMART CAMPUS MANAGEMENT")
print("====================================")

# Student records
students = ["Ravi", "Kiran", "Anil"]
attendance = [85, 72, 90]
marks = [[75, 80, 70], [65, 55, 60], [90, 85, 88]]
events = ["Sports", "Coding", "Cultural"]
registrations = [0, 0, 0]
resources = ["Computers", "Projectors", "Labs"]
stock = [20, 5, 3]

while True:
    print("\n======================================")
    print("             MAIN MENU")
    print("======================================")
    print("1. View Student Records")
    print("2. View Attendance")
    print("3. View Student Marks")
    print("4. Event Registration")
    print("5. Resource Allocation")
    print("6. Summary Report")
    print("7. Exit")

    choice = int(input("\nEnter your choice: "))

    if choice == 1:
        print("\nSTUDENT RECORDS")
        for i in range(len(students)):
            print(i + 1, ".", students[i])
```

```
elif choice == 2:
    print("\nATTENDANCE REPORT")
    for i in range(len(students)):
        print("\nStudent:", students[i])
        print("Attendance:", attendance[i], "%")
        if attendance[i] >= 75:
            print("Status: Eligible")
        else:
            print("Status: Shortage")

elif choice == 3:
    print("\nSTUDENT MARKS REPORT")
    for i in range(len(students)):
        total = 0
        for j in range(len(marks[i])):
            total += marks[i][j]
        average = total / len(marks[i])
        print("\nStudent:", students[i])
        print("Total Marks:", total)
        print("Average:", round(average, 2))

elif choice == 4:
    print("\nAVAILABLE EVENTS")
    for i in range(len(events)):
        print(i + 1, ".", events[i])
    event_choice = int(input("Select Event: "))
    if event_choice >= 1 and event_choice <= 3:
        registrations[event_choice - 1] += 1
        print("Registration Successful")
    else:
        print("Invalid Event Choice")

elif choice == 5:
    print("\nAVAILABLE RESOURCES")
    for i in range(len(resources)):
        print(i + 1, ".", resources[i],
              "- Available:", stock[i])
    resource_choice = int(input("Select Resource: "))
    quantity = int(input("Enter Quantity: "))
    if resource_choice >= 1 and resource_choice <= 3:
        index = resource_choice - 1
        if quantity > 0 and quantity <= stock[index]:
            stock[index] -= quantity
            print("Resource Allocated Successfully")
        else:
            print("Insufficient Stock or Invalid Quantity")
    else:
        print("Invalid Resource Choice")

elif choice == 6:
    print("\n======================================")
    print("          CAMPUS SUMMARY")
    print("======================================")
    print("Total Students:", len(students))
    eligible = 0
    for i in range(len(students)):
        if attendance[i] >= 75:
            eligible += 1
    print("Attendance Eligible Students:", eligible)
    print("\nEVENT REGISTRATIONS")
    for i in range(len(events)):
        print(events[i], ":", registrations[i])
    print("\nREMAINING RESOURCES")
    for i in range(len(resources)):
        print(resources[i], ":", stock[i])
elif choice == 7:
```

```
        print("\nThank You!")
        print("Exiting Smart Campus Management System")
        break
    else:
        print("Invalid Choice! Try Again.")
```

4. Sample Output (Example Menu Session)

Example selections: Attendance (2), Event Registration (4, Coding), Resource Allocation (5, Computers, quantity 5), Summary (6), Exit (7).

```
====================================
      SMART CAMPUS MANAGEMENT
====================================

====================================
             MAIN MENU
====================================
1. View Student Records
2. View Attendance
3. View Student Marks
4. Event Registration
5. Resource Allocation
6. Summary Report
7. Exit

Enter your choice: 2

ATTENDANCE REPORT

Student: Ravi
Attendance: 85 %
Status: Eligible

Student: Kiran
Attendance: 72 %
Status: Shortage

Student: Anil
Attendance: 90 %
Status: Eligible

====================================
             MAIN MENU
====================================
1. View Student Records
2. View Attendance
3. View Student Marks
4. Event Registration
5. Resource Allocation
6. Summary Report
7. Exit

Enter your choice: 4

AVAILABLE EVENTS
1 . Sports
2 . Coding
3 . Cultural
Select Event: 2
Registration Successful

====================================
             MAIN MENU
====================================
1. View Student Records
2. View Attendance
3. View Student Marks
4. Event Registration
5. Resource Allocation
6. Summary Report
```

```
7. Exit

Enter your choice: 5

AVAILABLE RESOURCES
1 . Computers - Available: 20
2 . Projectors - Available: 5
3 . Labs - Available: 3
Select Resource: 1
Enter Quantity: 5
Resource Allocated Successfully

=====================================
             MAIN MENU
=====================================
1. View Student Records
2. View Attendance
3. View Student Marks
4. Event Registration
5. Resource Allocation
6. Summary Report
7. Exit

Enter your choice: 6

=====================================
          CAMPUS SUMMARY
=====================================
Total Students: 3
Attendance Eligible Students: 2

EVENT REGISTRATIONS
Sports : 0
Coding : 1
Cultural : 0

REMAINING RESOURCES
Computers : 15
Projectors : 5
Labs : 3

=====================================
             MAIN MENU
=====================================
1. View Student Records
2. View Attendance
3. View Student Marks
4. Event Registration
5. Resource Allocation
6. Summary Report
7. Exit

Enter your choice: 7

Thank You!
Exiting Smart Campus Management System
```

5. Flowchart (Text Form)

```
START
  |
Initialize campus records
  |
Display menu and read choice
  |
Choice 1-3: View student data / attendance / marks
```

```
Choice 4: Register for event
Choice 5: Allocate available resource
Choice 6: Display campus summary
Choice 7: Exit -> STOP
Other choice: Show invalid choice
  |
Repeat menu unless Exit
```

6. Python Concepts Used

Variables, data types, operators, conditional statements, while and for loops, strings, lists, and nested lists. The provided program does not require tuples or tuple unpacking.

7. Result

The system successfully manages three sample students, checks attendance, calculates marks, registers campus events, allocates resources, and displays updated summary information. In the sample session, two students meet the attendance requirement, one Coding registration is recorded, and 15 computers remain after allocating five.